

PRACTICE SHEET 1

1. If $a : b = 7 : 9$ and $b : c = 15 : 7$, then what is $a : c$?
 (1) 5 : 3 (2) 3 : 5
 (3) 7 : 21 (4) 7 : 15
2. If $x = \frac{1}{3}y$ and $y = \frac{1}{2}z$, then $x : y : z$, is equal to:
 (1) 3 : 2 : 1 (2) 1 : 2 : 6
 (3) 1 : 3 : 6 (4) 2 : 4 : 6
3. If $x : y = 3 : 1$, then $x^3 - y^3 : x^3 + y^3 =$
 (1) 13 : 14 (2) 14 : 13
 (3) 10 : 11 (4) 11 : 10
4. The fourth proportional to 0.12, 0.21, 8 is :
 (1) 8.9 (2) 56
 (3) 14 (4) 17
5. The ratio $2^{1.5} : 2^{0.5}$ is the same as
 (1) 2 : 1 (2) 3 : 1
 (3) 6 : 1 (4) 3 : 2
6. If $A : B = 3 : 4$, $B : C = 5 : 7$ and $C : D = 8 : 9$ then $A : D$ is equal to
 (1) 3 : 7 (2) 7 : 3
 (3) 21 : 10 (4) 10 : 21
7. If b is the mean proportional of a and c , then $(a - b)^3 : (b - c)^3$ equals
 (1) $a^3 : c^3$ (2) $b^2 : c^2$
 (3) $a^2 : c^2$ (4) $a^3 : b^3$
8. If $a : b = 5 : 7$ and $c : d = 2a : 3b$, then $ac : bd$ is :
 (1) 20 : 38 (2) 50 : 147
 (3) 10 : 21 (4) 50 : 151
9. If $A : B : C = 2 : 3 : 4$, then ratio $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$ is equal to
 (1) 8 : 9 : 16 (2) 8 : 9 : 12
 (3) 8 : 9 : 24 (4) 4 : 9 : 16
10. If $a : b = c : d = e : f = 1 : 2$, then $(3a + 5c + 7e) : (3b + 5d + 7f)$ is equal to
 (1) 8 : 7 (2) 2 : 1
 (3) 1 : 4 (4) 1 : 2
11. The mean proportional between $(3 + \sqrt{2})$ and $(12 - \sqrt{32})$ is
 (1) 7 (2) $2\sqrt{7}$
 (3) 6 (4) $\frac{15 - 3\sqrt{2}}{2}$
12. If $a : b : c = 2 : 3 : 4$ and $2a - 3b + 4c = 33$, then the value of c is
 (1) 6 (2) 9
 (3) 12 (4) $66/7$
13. If two times of A is equal to three times of B and also equal to four times of C , then $A : B : C$ is
 (1) 2 : 3 : 4 (2) 3 : 4 : 2
 (3) 4 : 6 : 3 (4) 6 : 4 : 3
14. If A and B are in the ratio 3 : 4, and B and C in the ratio 12 : 13, then A and C will be in the ratio
 (1) 3 : 13 (2) 9 : 13
 (3) 36 : 13 (4) 13 : 9
15. If $3x = 5y = 4z$, then $x : y : z$ is equal to
 (1) 9 : 12 : 16 (2) 20 : 12 : 15
 (3) 15 : 10 : 9 (4) 8 : 5 : 3
16. If $A : B = 3 : 4$ and $B : C = 6 : 5$, then $A : (A + C)$ is equal to
 (1) 9 : 10 (2) 10 : 9
 (3) 9 : 19 (4) 19 : 9
17. If 78 is divided into three parts which are in the ratio $1 : \frac{1}{3} : \frac{1}{6}$, the middle part is
 (1) $9\frac{1}{3}$ (2) 13
 (3) $17\frac{1}{3}$ (4) $18\frac{1}{3}$
18. If two-third of A is four-fifth of B , then $A : B = ?$
 (1) 5 : 6 (2) 6 : 5
 (3) 10 : 9 (4) 9 : 10
19. If $A = \frac{4}{5}$ of B and $B = \frac{5}{2}$ of C , then the ratio of $A : C$ is
 (1) 1 : 2 (2) 2 : 1
 (3) 2 : 3 (4) 1 : 3
20. Marks of two candidates P and Q are in the ratio of 2 : 5. If the marks of P is 120, marks of Q will be

(1) 120
(3) 300

(2) 240
(4) 360

21. If $A : B = 4 : 9$ and $A : C = 2 : 3$, then $(A + B) : (A + C)$ is

(1) 15 : 13
(3) 13 : 10

(2) 10 : 13
(4) 13 : 15

22. If $(a + b) : (b + c) : (c + a) = 6 : 7 : 8$ and $(a + b + c) = 14$, then the value of c is

(1) 6
(3) 8

(2) 7
(4) 14

23. If 18, x and 50 are in continued proportion, then the value of x is

(1) 30
(3) 5

(2) 3
(4) 32

24. If $(a + b) : \sqrt{ab} = 4 : 1$, where $a > b > 0$, then $a : b$ is

(1) $(2 + \sqrt{3}) : (2 - \sqrt{3})$
(2) $(2 - \sqrt{3}) : (2 + \sqrt{3})$
(3) $(3 + \sqrt{2}) : (3 - \sqrt{2})$
(4) $(3 - \sqrt{2}) : (3 + \sqrt{2})$

25. Rs. 2420 were divided among A, B and C so that $A : B = 5 : 4$ and $B : C = 9 : 10$ then C gets

(1) Rs. 680
(3) Rs. 900

(2) Rs. 800
(4) Rs. 950

26. If $A = \frac{1}{4}B$ and $B = \frac{1}{2}C$, then $A : B : C$ is :

(1) 8 : 4 : 1
(3) 1 : 4 : 8

(2) 4 : 2 : 1
(4) 1 : 2 : 4

27. If $2A = 3B = 4C$, then $A : B : C$ is :

(1) 2 : 3 : 4
(3) 6 : 4 : 3

(2) 4 : 3 : 2
(4) 3 : 4 : 6

28. The ratio $4^{3.5} : 2^5$ is the same as

(1) 4 : 1
(3) 1 : 2

(2) 2 : 1
(4) 1 : 4

29. There are three numbers A, B, C such that twice A is equal to thrice B and four times B is equal to five times C. Then the ratio between A and C is

(1) 3 : 4
(3) 15 : 8

(2) 8 : 15
(4) 4 : 3

30. If $\frac{3x+5}{5x-2} = \frac{2}{3}$, then the value of x is

(1) 11
(3) 23

(2) 19
(4) 7

ANSWER KEY

1.	(1)	2.	(3)	3.	(1)	4.	(3)	5.	(1)	6.	(4)	7.	(4)	8.	(2)	9.	(3)	10.	(4)
11.	(2)	12.	(3)	13.	(4)	14.	(2)	15.	(2)	16.	(3)	17.	(3)	18.	(2)	19.	(2)	20.	(3)
21.	(3)	22.	(1)	23.	(1)	24.	(1)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(3)	30.	(2)

PRACTICE SHEET 2

1. Ram, Shyam and Kamal together started a business in partnership. The ratio of their capitals is 3:4:7. If their annual profit be Rs. 21000, what will be Kamal's share in this profit?
 (1) Rs. 12500 (b) Rs. 10500
 (3) Rs. 15000 (4) Rs. 10000
2. Three partners A,B,C in a business invested money such that 6(A's capital)=8 (B's capital)=10 (C's capital) then the ratio of their capitals is:
 (1) 3:4:5 (2) 12:15:20
 (3) 20:15:12 (4) 6:8:10
3. If 4 (A's capital) = 6 (B's capital) = 10 (C's capital), then out of a profit of Rs. 4650, C will receive
 (1) 465 (2) 900
 (3) 1550 (4) 2250
4. Three partners A,B,C in a business invested money such that (A's capital)/3 = (B's capital)/1 = [C's capital]/5 then the ratio of their capitals is:
 (1) 3:1:5 (2) 12:15:20
 (3) 20:15:12 (4) 6:8:10
5. In a business A and b invested amounts in ratio 2:3, where as B and C invested amounts in the ration 3:4. If their annual profit be 45000 then B's share in profit is:
 (1) Rs. 15000 (2) Rs. 30000
 (3) Rs. 20000 (4) None
6. A,B and C invested Rs. 26000, Rs. 34000 and Rs. 10000 respectively in a business. At the end of the year, they earn a profit of Rs.3500. B's share of profit is:
 (1) Rs. 1200 (2) Rs. 1500
 (3) Rs. 1700 (4) Rs. 1900
7. Four milkmen rented a pasture A grazed 18 cows for 4 months B 25 cows for 2 months C 28 cows for 5 months and D 21 cows for 3 months. total rent of the field is 1625, A's share of rent is:
 (1) Rs. 250 (2) Rs. 300
 (3) Rs. 360 (4) Rs. 650
8. A,B,C rented a video cassette at a rent of 350. If they use it for 6 hrs, 10 hrs and 12 hrs. Rent to be paid C, is
 (1) Rs. 75 (2) Rs. 125
 (3) Rs. 135 (4) Rs. 150
9. Amit started a business investing Rs. 25000. After 3 months, Vinay joined him. With a capital of Rs. 30000. At the end of the year they make a profit of Rs. 19000. What will be Amit's share in the profit?
 (1) Rs. 9423 (2) Rs. 10000
 (3) Rs. 12500 (4) none
10. A,B and C start a business each investing Rs. 20000. After 5 months A withdrew Rs. 5000, B Rs. 4000 and C invested Rs. 6000 more. At the end of the year a total profit of Rs. 69900 was recorded. What is share of B?
 (1) Rs. 20500 (2) Rs. 21200
 (3) Rs. 27300 (4) Rs. 28200
11. A,B,C enter into partnership. A initially invests Rs. 25 lakhs and another Rs.10 Lakhs after one year. B initially invests Rs. 35 lakhs and withdraws Rs. 10 lakhs after 2 years and C invests Rs. 30 lakhs. In what ratio should the profits be divided at the end of 3 years?
 (1) 10:10:9 (2) 19:19:18
 (3) 20:20:19 (4) None
12. A,BC enter into partnership and their capitals are in the proportion 1/3:1/4:1/5, a withdraws half his capital after 4 months. Out of a total annual profit of Rs. 8470, A's share is:
 (1) Rs. 2520 (2) Rs. 2800
 (3) Rs. 3150 (4) Rs. 4120
13. In a partnership A invests 1/6 of the capital of 1/6 of the time B invests 1/3 of the capital for 1/3 of the time and C, the rest of the capital for the Whole time. Out of a profit of Rs. 23000, B's share is
 (1) Rs. 4000 (2) Rs. 5000
 (3) Rs. 3250 (4) Rs. 4800
14. A and B started a business with initial investments in ratio 12:11 and their annual profits were in the ratio 4:1. If A invested the

money for 11 months B invested the money for

- (1) 3 months (2) 4 months
(3) 5 months (4) 7 months

15. A began a business with Rs. 4500 and was joined afterwards by B with Rs. 5400. If the profits at the end of the year were divided in the ratio 2:1, B joined the business after:

- (1) 4 month (2) 5 month
(3) 6 month (4) 7 month

16. A and B partners in a business. A contributes $\frac{1}{4}$ of the capital for 15 months and B received $\frac{2}{3}$ of the profit. Find how long B's money was used.

- (1) 6 month (2) 9 month
(3) 10 month (4) 1 year

17. A and B enter into partnership. A Invests Rs. 16000 for 8 months and B remains in the business for 4 months. Out off a total profit, B claims $\frac{2}{7}$ of the profit. How much money was contributed by B?

- (1) 11900 (2) 10500
(3) 13600 (4) 12800

18. A is a working and B, a sleeping partner in a business. A puts in Rs. 12000 and B Rs. 20000. A receives 10% of the profits for managing, the rest being divided in proportion to their capitals. Out of a total profit of Rs. 18000 the money received by A is:

- (1) Rs. 6480 (2) Rs. 8400
(3) Rs. 7875 (4) Rs.8325

19. A is a working and B, a sleeping partner in a business. A puts in Rs. 20000 and B Rs. 30000. A receives 20% of the profits for managing, the rest being divided in proportion 'to their capitals. Out of total profit of Rs. 50000 the money received by A is:

- (1) Rs. 26000 (2) Rs. 24000
(3) Rs. 30000 (4) Rs.20000

20. A,B,C subscribe Rs. 50000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 More than C. Out of a total profit of Rs. 35000, A receives:

- (1) 8400 (2) 11900
(3) 13600 (4) 14700

21. A,B,C subscribe Rs. 80000 for a business. A subscribes Rs. 10000 more than B and B

Rs. 20000 More than C. Out of a total profit of Rs. 40000, A receives:

- (1) 32000 (2) 31900
(3) 33600 (4) 20000

22. A,B and C enter into partnership. A invests some money at the begging B invests double the amount after 6 months and C invests thrice the amount after 8 months. If the annual profit be Rs. 27000, C's share is:

- (1) 8625 (2) 9000
(3) 10800 (4) 11250

23. A, B and C entered in to a partnership. A invested Rs. 2560 and B invested Rs. 2000. At the end of the year they gained Rs. 1105, out of which a got Rs. 320. C's capital was:

- (1) Rs. 2840 (2) Rs. 4028
(3) Rs. 4280 (4) Rs. 4820

24. A,B and Care three partners in a business. If A's capital is equal to twice B's capital and B 's capital is three times C's capital, ratio of the capitals of A,B,C is

- (1) 2:1:3 (2) 1:2:6
(3) 6:3:1 (4) 1:3:6

25. A,B and C start a business. A invests 3 times as much as B. invests and B invests two third of C invests. Then the ratio of capitals of A,B ,C is:

- (1) 3:9:2 (2) 6:10:15
(3) 5:3:2 (4) 6:2:3

26. A,B,C started a shop by investing Rs.27000, Rs. 81000 and Rs.72000 respectively. At end of the one year, B's share of total profit was Rs.36000. What total profit?

- (1) 108000 (2) 116000
(3) 80000 (4) 92000

27. A and B started a joint firm. A's investment was thrice the investment of B and the period of his investment was twice the period of investment of B. If B got Rs. 12000 as profit then the total profit was

- (1) 72000 (2) 48000
(3) 84000 (4) 60000

28. A and b entered into partnership with capitals in the ratio 4:5. After 3 months A withdraw $\frac{1}{4}$ of his capital and B. withdrew $\frac{1}{5}$ of this capital. The gain at the end of 10 months was Rs. 15200. A's share in this profit is:

(1) Rs. 7200
(3) Rs. 7800

(2) Rs. 6600
(4) Rs. 8200

29. A,B,C became partners in a business by investing money in ratio 5:7:6. Next year they increase their investments by 26%, 20% and 15%. profit earned during 2nd year should be distributed in ratio.

(1) 21:28:23
(3) 31:27:21

(2) 26:20:15
(4) None

30. A and B invest in business in tlhe ratio 3:2. If 5% of the total profit goes to charity and A's share is Rs. 8550, then total profit is:
(1) 15760
(2) 15735
(3) 15000
(4) 142500

ANSWER KEY

1.	(2)	2.	(3)	3.	(2)	4.	(1)	5.	(1)	6.	(3)	7.	(3)	8.	(4)	9.	(2)	10.	(2)
11.	(2)	12.	(2)	13.	(1)	14.	(1)	15.	(4)	16.	(3)	17.	(4)	18.	(3)	19.	(1)	20.	(4)
21.	(4)	22.	(2)	23.	(3)	24.	(3)	25.	(4)	26.	(3)	27.	(3)	28.	(2)	29.	(1)	30.	(3)

PRACTICE SHEET 3

- The ratio of the age of Ram and Rahim 10 years ago was 1 : 3. The ratio of their age five years hence will be 2 : 3. Then the ratio of their present age is
(1) 1 : 2
(2) 3 : 5
(3) 3 : 4
(4) 2 : 5
- The ratio of the age of a father to that of his son is 5: 2. If the product of their ages in years is 1000, then the father's age (in years) after 10 years will be :
(1) 50
(2) 60
(3) 80
(4) 100
- The ratio between Sumit's and Prakash's age at present is 2 : 3. Sumit is 6 years younger than Prakash. The ratio of Sumit's age to Prakash's age after 6 years will be
(1) 2 : 3
(2) 1 : 2
(3) 4 : 3
(4) 3 : 4
- Harsha is 40 years old and Ritu is 60 years old. How many years ago was the ratio of their ages 3 : 5 ?
(1) 10 years
(2) 20 years
(3) 37 years
(4) 5 years
- The ratio of present age of two brothers is 1 : 2 and 5 years back the ratio was 1 : 3. What will be the ratio of their age after 5 years ?
(1) 1 : 4
(2) 2 : 3
(3) 3 : 5
(4) 5 : 6
- Four years ago, the ratio of the age of A and B was 2 : 3 and after four years it will become 5 : 7. Find their present age.
(1) 36 years and 40 years
(2) 32 years and 48 years
(3) 40 years and 56 years
(4) 36 years and 52 years
- The average age of boys in the class is twice the number of girls in the class. The ratio of boys and girls in the class of 50 is 4 : 1. The total of the ages (in years) of the boys in the class is
(1) 2000
(2) 2500
(3) 800
(4) 400
- The ratio of age of two boys is 5 : 6. After two years the ratio will be 7 : 8. The ratio of their age after 12 years will be
(1) 22/24
(2) 15/16
(3) 17/18
(4) 11/12
- The ratio of the present age of Puneet and Appu is 2 : 3. After 3 years the ratio of their age will be 3 : 4. The present age of Puneet is:
(1) 3 years
(2) 6 years
(3) 9 years
(4) 4 years
- The ratio of the ages of a father and his son 10 years hence will be 5 : 3, while 10 years ago, it was 3:1. The ratio of the age of the son to that of the father today, is

- (1) 1 : 2 (2) 1 : 3
(3) 2 : 3 (4) 2 : 5
- 11.** Father is three times more than his son Ronit. After 8 years, he would be two and half times of Ronit's age, after further 16 years, how many times would he be of Ronit's age
(1) 2 times (b) 2 and half times
(3) 2.75 times (4) 3 times
- 12.** The sum of ages of 5 children born at the intervals of 3 years each is 50 years what is the age of the youngest child
(1) 4 years (2) 8 years
(3) 10 years (4) none of these
- 13.** A father said to his son "I was as old as you at the present at the time of your birth" if the father's age is 38 years now the son's age 5 years back was
(a) 14 years (b) 19 years
(c) 33 years (d) 38 years
- 14.** A is 2 years older than B who is twice as old as C. If the total of the ages of A, B and C to be 27, then how old is B
(a) 7 (b) 8
(c) 9 (d) 10
- 15.** Present age of Sameer and Anand are in the ratio 5 : 4 respectively. 3 years hence, the ratio of their ages will become 11:9 respectively. what is Anand's present age in years
(1) 24 (2) 27
(3) 40 (4) 50
- 16.** A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is
(1) 14 years (2) 18 years
(3) 20 years (4) 22 years
- 17.** Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5 . four years hence the ratio of their ages will be 11 :10 . what is present age of Sagar
(1) 16 years (2) 18 years
(3) 20 years (4) 22 years
- 18.** The sum of present age of father and his son is 60 years. Six years ago, father's age was five times the age of his son, after 6 years son's age will be
(1) 12 years (2) 14 years
(3) 18 years (4) 20 years
- 19.** At present the ratio between the ages of Arun and Deepak is 4:3, After 6 years, Arun's age will be 26 years. what is the age of Deepak at present
(1) 12 years (2) 15 years
(3) 19 and half (4) 21 years
- 20.** Sachin is younger than Rahul by 7 years. if their ages are in the respective ratio of 7:9, then how old is Sachin
(1) 16 years (2) 18 years
(3) 28 years (4) 24.5 years
- 21.** The present age of three persons in proportions 4 : 7 : 9 , Eight years ago the sum of their ages was 56. find their present age
(1) 8, 20, 28 (2) 16, 28, 36
(3) 20, 35, 45 (4) none of these
- 22.** Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother was 4 years younger to her was born what is the difference between the ages of his parents
(1) 2 years (2) 4 years
(3) 6 years (4) 8 years
- 23.** Rs 180 contained in a box consists of one rupee, 50 paise and 25 paise coins in the ratio 2 : 3 : 4. What is the number of 50 paise coins?
(1) 60 (2) 120
(3) 150 (4) 180
- 24.** If 378 coins consist of rupees, 50 paise and 25 paise coins, whose values are in the ratio of 13 : 11 : 7, the number of 50 paise coins will be :
(1) 132 (2) 128
(3) 136 (4) 133
- 25.** A bag contains Rs 90 in coins of denominations of 50 paise, 25 paise and 10 paise. If coins of 50 paise, 25 paise and 10 paise are in the ratio 2 : 3 : 5, then the number of 25 paise coins in the bag is
(1) 80 (2) 120
(3) 100 (4) 135
- 26.** There are Rs 225 consisting of one rupee, 50 paise and 25 paise coins. The ratio of their numbers in that order is 8 : 5 : 3. The number of one-rupee coins is :
(1) 80 (2) 112
(3) 160 (4) 172

- 27.** A box contains 1-rupee, 50-paise and 25-paise coins in the ratio 8 : 5 : 3. If the total amount of money in the box is 112.50, the number of 50-paise coins is
 (1) 80 (2) 50
 (3) 30 (4) 42
- 28.** In a bag, there are three types of coins — 1-rupee, 50 paise and 25-paise in the ratio of 3 : 8 : 20. Their total value is Rs 372. The total number of coins is
 (1) 1200 (2) 961
 (3) 744 (4) 612
- 29.** A box has 210 coins of denominations one-rupee and fifty paise only. The ratio of their respective values is 13 : 11. The number of one-rupee coins is
 (1) 65 (2) 66
 (3) 77 (4) 78
- 30.** A boy has a few coins of denominations 50 paise, 25 paise and 10 paise in the ratio 1 : 2 : 3. If the total amount of the coins is 6.50, the number of 10 paise coins is
 (1) 5 (2) 10
 (3) 15 (4) 20

ANSWER KEY

1.	(2)	2.	(2)	3.	(4)	4.	(1)	5.	(3)	6.	(4)	7.	(3)	8.	(3)	9.	(2)	10.	(1)
11.	(1)	12.	(1)	13.	(1)	14.	(4)	15.	(1)	16.	(4)	17.	(1)	18.	(4)	19.	(2)	20.	(4)
21.	(2)	22.	(3)	23.	(2)	24.	(1)	25.	(2)	26.	(3)	27.	(2)	28.	(2)	29.	(4)	30.	(3)

PRACTICE SHEET 4

- 1.** The income of A, B and C are in the ratio 3 : 7 : 4 and their expenses in the ratio 4 : 3 : 5. If A saves 300 out of an income of 2,400, the savings of B and C are :
 (1) 4025 and 575 (2) 1575 and 2.625
 (3) 2750 and 1.525 (4) 3725 and 1.525
- 2.** Between two consecutive years my income are in the ratio of 2 : 3 and expenses in the ratio 5 : 9. If my income in the second year is 45000 and my expenses in the first year is 25000 my total savings for the two years is :
 (1) Nil (2) 15000
 (3) 10000 (4) 5000
- 3.** A and B have monthly incomes in the ratio 5 : 6 and monthly expenditures in the ratio 3 : 4. If they save 1800 and 1600 respectively, find the monthly income of B :
 (1) 3400 (2) 2700
 (3) 1720 (4) 7200
- 4.** The ratio of income of two persons is 5 : 3 and that of their expenditures is 9 : 5. Find the income of each person, if they save 1,300 and 900 respectively.
 (1) 4,000, 2,400 (2) 3,000, 1,800
 (3) 5,000, 3,000 (4) 4,500, 2,700
- 5.** The annual income of A and B are in the ratio 4 : 3 and the ratio of their expenditures is 3 : 2. If each of them saves 600 in the year, the annual income of A is
 (1) 4800 (2) 1800
 (3) 1200 (4) 2400
- 6.** The income of A, B and C are in the ratio 7 : 9 : 12 and their spending are in the ratio 8 : 9 : 15. If A saves $\frac{1}{4}$ th of his income, then the savings of A, B and C are in the ratio of :
 (1) 56 : 99 : 69 (2) 69 : 56 : 99
 (3) 99 : 56 : 69 (4) 99 : 69 : 56
- 7.** The ratio of income of P and Q is 3 : 4 and the ratio of their expenditures is 2 : 3. If both of them save 6000, the income of P is
 (1) 20000 (2) 12000
 (3) 18000 (4) 24000
- 8.** A man spends a part of his monthly income and saves a part of it. The ratio of his expenditure to his saving is 26 : 3. If his

monthly income is 7250, what is the amount of his monthly savings ?

- (1) 350 (2) 290
(3) 750 (4) 780

9. The monthly salaries of A, B and C are in the ratio 2 : 3 : 5. If C's monthly salary is 12,000 more than that of A, then B's annual salary is
(1) 1,20,000 (2) 1,44,000
(3) 1,80,000 (4) 2,40,000

10. The ratio of income of two persons is 5 : 3 and that of their expenditures is 9 : 5. If they save 2600 and 1800 respectively, their incomes are :
(1) 8000; 4800 (2) 6000; 3600
(3) 10000; 6000 (4) 9000; 5400

11. The monthly income of two persons are in the ratio 2 : 3 and their monthly expenses are in the ratio 5 : 9. If each of them saves 600 per month, then their monthly incomes are
(1) 1,500; 2,250 (2) 1,200; 1,800
(3) 1,600; 2,400 (4) 1,400; 2,100

12. A person bought some rice and wheat for 380. The ratio of weight of rice and wheat is 4 : 3 and the price of equal amount of rice and wheat is in the ratio 5 : 6. The rice was bought of worth
(1) 380 (2) 300
(3) 200 (4) 180

13. Divide 7,500 among A, B and C such that A's share to B's share is in ratio 5 : 2 and B's share to C's share is in the ratio 7 : 13. How much will B receive?
(1) 1,400 (2) 3,500
(3) 2,600 (4) 7,000

14. A sum of 1240 is distributed among A, B and C such that the ratio of amount received by A and B is 6 : 5 and that of B and C is 10 : 9 respectively. Find the share of C.
(1) 480 (2) 360
(3) 400 (4) 630

15. 3400 is divided among A, B, C, D in such a way that the share of A and B, B and C, C and D may be as 2 : 3, 4 : 3 and 2 : 3 respectively. The sum of shares of B and D is
(1) 2040 (2) 1680
(3) 2000 (4) 1720

16. 750 are divided among A, B and C in such a manner that A : B is 5 : 2 and B : C is 7 : 13. What is A's share ?

- (1) 350 (2) 260
(3) 140 (4) 250

17. In an army selection process, the ratio of selected to unselected candidates was 3 : 1. If 80 less had applied and 40 less selected, the ratio of selected to unselected candidates would have been 4 : 1. How many candidates had applied for the process?
(1) 480 (2) 960
(3) 240 (4) 1440

18. In an army selection process, the ratio of selected to unselected candidates was 4:1. If 90 less had applied and 20 less were selected, the ratio of selected to unselected candidates would have been 5:1. How many candidates had applied for the process ?
(1) 1650 (2) 3300
(3) 825 (4) 4950

19. In a cricket match there are three types of tickets say A, B and C each costing Rs. 1000, Rs. 500 and Rs. 200 respectively. The ratio of the tickets sold of categories A, B and C is 3 : 2 : 5. If the total collection from selling the tickets is Rs 2.5 crore, find the total number of tickets sold?
(1) 5000 (2) 4800
(3) 50000 (4) 52000

20. From each of the two given unequal numbers, half the smaller number is subtracted. Then, of the resulting numbers, the larger one is five times than the smaller one. Then the ratio of the larger to smaller one is
(1) 2 : 1 (2) 3 : 2
(3) 3 : 1 (4) 1 : 4

21. A person ordered 4 shirts of brand A and some shirts of brand B. The price of one shirt of brand A was twice that of brand B. When the order was executed, it was found that the numbers of the two brands has been interchanged. This increased the bill by 40%. The ratio of the number of brand A shirts to that of brand B shirts in the original order was
(1) 1 : 2 (2) 1 : 3
(3) 1 : 4 (4) 1 : 5

22. A and B together have 158. C has 101 less than what A and B together have, and B has 23 more than C. The amount of A is :
(1) 80 (2) 78
(3) 57 (4) 88

- 23.** A sum of 340.68 is distributed among L, M and N such that L gets 5.72 more than N and M gets Rs. 2.24 more than L. N gets
 (1) 109 (2) 110.90
 (3) 113.56 (4) 114.72
- 24.** The ratio of the first and second class fares between two railway stations is 4 : 1 and that of the number of passengers travelling by first and second classes is 1 : 40. If on a day 1,100 are collected as total fare, the amount collected from the first class passengers is
 (1) 315 (2) 275
 (3) 137.50 (4) 100
- 25.** Three persons walk from place A to place B. Their speeds are in the ratio 4 : 3 : 5. The ratio of the time taken by them to reach B will be :
 (1) 10 : 15 : 13 (2) 2 : 3 : 4
 (3) 15 : 20 : 12 (4) 16 : 18 : 15
- 26.** Two numbers are in the ratio of 3 : 5. If 9 be subtracted from each, then they are in the ratio of 12 : 23. Find the numbers.
 (1) 15, 28 (2) 36, 115
 (3) 33, 55 (4) 60, 69
- 27.** Three numbers are in the ratio 1 : 2 : 3. By adding 5 to each of them, the new numbers are in the ratio 2 : 3 : 4. The numbers are:
 (1) 10, 20, 30 (2) 15, 30, 45
 (3) 1, 2, 3 (4) 5, 10, 15
- 28.** Ram got twice as many marks in English as in Science. His total marks in English, Science and Maths are 180. If the ratio of his marks in English and Maths is 2 : 3, what is his marks in Science ?
 (1) 30 (2) 60
 (3) 72 (4) 90
- 29.** The ratio of number of boys to the number of girls in a school of 432 pupils is 5 : 4. When some new boys and girls are admitted, the number of boys increase by 12 and the ratio of the boys to girls changes to 7 : 6. The number of new girls admitted is
 (1) 12 (2) 14
 (3) 24 (4) 20
- 30.** If the ratio of two numbers is 1 : 5 and their product is 320, then the difference between the squares of these two numbers is :
 (1) 1024 (2) 1256
 (3) 1536 (4) 1435

ANSWER KEY

1.	(1)	2.	(4)	3.	(4)	4.	(1)	5.	(4)	6.	(1)	7.	(3)	8.	(3)	9.	(2)	10.	(1)
11.	(3)	12.	(3)	13.	(1)	14.	(2)	15.	(1)	16.	(1)	17.	(1)	18.	(1)	19.	(3)	20.	(3)
21.	(2)	22.	(2)	23.	(1)	24.	(4)	25.	(3)	26.	(3)	27.	(4)	28.	(1)	29.	(3)	30.	(3)

PRACTICE SHEET 5

- 1.** A mixture of 30 litres contain milk and water in the ratio of 7 : 3. How much water should be added to it so that the ratio of milk and water becomes 3 : 7 ?
 (1) 40 litres (2) 49 litres
 (3) 56 litres (4) 63 litres
- 2.** A barrel contains a mixture of wine and water in the ratio 3 : 1. How much fraction of the mixture must be drawn off and substituted by water so that the ratio of wine and water in the resultant mixture in the barrel becomes 1 : 1 ?
 (1) $\frac{1}{4}$ (2) $\frac{1}{30}$
 (3) $\frac{3}{4}$ (4) $\frac{2}{3}$
- 3.** There is 81 litres pure milk in a container. One-third of milk is replaced by water in the container. Again one-third of mixture is extracted and equal amount of water is added. What is the ratio of milk to water in the new mixture?
 (1) 1 : 2 (2) 1 : 1
 (3) 2 : 1 (4) 4 : 5
- 4.** In 80 litres mixture of milk and water the ratio of amount of milk to that of amount of water is 7 : 3. In order to make this ratio 2 : 1, how many litres of water should be added?
 (1) 5 (2) 6
 (3) 8 (4) 4

5. Vessels A and B contain mixtures of milk and water in the ratios 4 : 5 and 5 : 1 respectively. In what ratio should quantities of mixture be taken from A and B to form a mixture in which milk to water is in the ratio 5 : 4?
- (1) 2 : 5 (2) 4 : 3
(3) 5 : 2 (4) 2 : 3
6. The milk and water in a mixture are in the ratio 7 : 5. When 15 litres of water are added to it, the ratio of milk and water in the new mixture becomes 7 : 8. The total quantity of water in the new mixture is
- (1) 35 litres (2) 40 litres
(3) 60 litres (4) 96 litres
7. In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7: 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1: 2 is
- (a) 30 (b) 32
(c) 33 (d) 35
8. Two equal vessels are filled with the mixtures of water and milk in the ratio of 3:4 and 5:3 respectively. If the mixtures are poured into a third vessel, the ratio of water and milk in the third vessel will be
- (1) 15 : 12 (2) 53 : 59
(3) 20 : 9 (4) 59 : 53
9. A Can contains a mixture of two liquids A and B in the ratio 7:5. When 9 litres of mixture are drained off and the Can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the Can initially ?
- (1) 10 litres (2) 20 litres
(3) 21 litres (4) 25 litres
10. The ratio of milk and water in mixtures of four containers are 5 : 3, 2 : 1, 3 : 2 and 7 : 4 respectively. In which container is the quantity of milk, relative to water, minimum?
- (1) First (2) Second
(3) Third (4) Fourth
11. An alloy contains copper, zinc and nickel in the ratio of 5 : 3 : 2. The quantity of nickel (in kg) that must be added to 100 kg of this alloy to have the new ratio 5 : 3 : 3 is
- (1) 8 (2) 10
(3) 12 (4) 15
12. In an alloy, zinc and copper are in the ratio 1 : 2. In the second alloy, the same elements are in the ratio 2 : 3. If these two alloys be mixed to form a new alloy in which two elements are in the ratio 5 : 8, the ratio of these two alloys in the new alloy is
- (1) 3 : 10 (2) 3 : 7
(3) 10 : 3 (4) 7 : 3
13. A liquid 'P' is $1\frac{3}{7}$ times as heavy as water and water is $1\frac{2}{5}$ times as heavy as another liquid 'Q'. The amount of liquid 'P' that must be added to 7 litres of the liquid 'Q' so that the mixture may weigh as much as an equal volume of water, will be
- (1) 7 litres (2) $5\frac{1}{6}$ litres
(3) 5 litres (4) $4\frac{2}{3}$ litres
14. Two alloys contain tin and iron in the ratio of 1 : 2 and 2 : 3. If the two alloys are mixed in the proportion of 3 : 4 respectively (by weight), the ratio of tin and iron in the newly formed alloy is :
- (1) 10 : 21 (2) 13 : 22
(3) 14 : 25 (4) 12 : 23
15. 60 kg of an alloy A is mixed with 100 kg of alloy B. If alloy A has lead and tin in the ratio 3 : 2 and alloy B has tin and copper in the ratio 1 : 4, the amount of tin in the new alloy is
- (1) 53 kg (2) 44 kg
(3) 80 kg (4) 24 kg
16. Three glasses of equal volume contains acid mixed with water. The ratios of acid and water are 2 : 3, 3 : 4 and 4 : 5 respectively. Contents of these glasses are poured in a large vessel. The ratio of acid and water in the large vessel is
- (1) 411 : 540 (2) 401 : 544
(3) 417 : 564 (4) 407 : 560
17. Zinc and copper are in the ratio of 5 : 3 in 200 gm of an alloy. How much grams of copper be added to make the ratio as 3 : 5?
- (1) 400/3 (2) 1/200
(3) 72 (4) 66

- 18.** The ratio of copper and zinc in brass is 13 : 7. How much zinc will be there in 100 kg of brass ?
 (1) 20 kg. (2) 55 kg.
 (3) 35 kg. (4) 40 kg.
- 19.** In 30 litres mixture of acid, the ratio of acid and water is 2 : 3. What amount of water should be added to the mixture so that the ratio of acid and water becomes 2 : 5 ?
 (1) 10 litres (2) 15 litres
 (3) 18 litres (4) 12 litres
- 20.** There are three containers of equal capacity. The ratio of Sulphuric acid to water in the first container is 3 : 2, that in the second container is 7 : 3 and in the third container it is 11 : 4. If all the liquids are mixed together, then the ratio of Sulphuric acid to water in the mixture will be :
 (1) 61 : 29 (2) 61 : 28
 (3) 60 : 29 (4) 59 : 29
- 21.** 200 litres of a mixture contains milk and water in the ratio 17 : 3. After the addition of some more milk to it, the ratio of milk to water in the resulting mixture becomes 7 : 1. The quantity of milk added to it was
 (1) 20 litres (2) 40 litres
 (3) 60 litres (4) 80 litres
- 22.** The ratio in which a man must mix rice at 10.20 per kg and 14.40 per kg so as to make a mixture worth 12.60 per kg, is
 (1) 4 : 3 (2) 2 : 5
 (2) 18 : 24 (4) 3 : 4
- 23.** A mixture contains spirit and water in the ratio 3 : 2. If it contains 3 litres more spirit than water, the quantity of spirit in the mixture is
 (1) 10 litres (2) 12 litres
 (3) 8 litres (4) 9 litres
- 24.** A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup ?
 (1) 1/3 (2) 1/4
- (3) 1/5 (4) 1/7
- 25.** Two vessels A and B contain milk and water mixed in the ratio 4 : 3 and 2 : 3. The ratio in which these mixtures be mixed to form a new mixture containing half milk and half water is
 (1) 7 : 5 (2) 6 : 5
 (3) 5 : 6 (4) 4 : 3
- 26.** A mixture contains alcohol and water in the ratio 4 : 3. If 5 litres of water is added to the mixture, the ratio becomes 4 : 5. The quantity of alcohol in the given mixture is
 (1) 3 litres (2) 4 litres
 (3) 15 litres (4) 10 litres
- 27.** In two alloys A and B, the ratio of zinc to tin is 5 : 2 and 3 : 4 respectively. Seven kg of the alloy A and 21 kg of the alloy B are mixed together to form a new alloy. What will be the ratio of zinc and tin in the new alloy ?
 (1) 2 : 1 (2) 1 : 2
 (3) 2 : 3 (4) 1 : 1
- 28.** A jar contained a mixture of two liquids A and B in the ratio 4 : 1. When 10 litres of the mixture was taken out and 10 litres of liquid B was poured into the jar, this ratio became 2 : 3. The quantity of liquid A contained in the jar initially was
 (1) 4 litres (2) 8 litres
 (3) 16 litres (4) 40 litres
- 29.** A vessel contains 20 litres of acid. 4 litres of acid is taken out of the vessel and replaced by the same quantity of water. Next 4 litres of the mixture are withdrawn, and again the vessel is filled with the same quantity of acid left in the vessel with the quantity of acid initially in the vessel is
 (1) 4 : 5 (2) 4 : 25
 (3) 16 : 25 (4) 1 : 5
- 30.** 729 ml of a mixture contains milk and water in the ratio 7 : 2. How much more water is to be added to get a new mixture containing milk and water in the ratio 7 : 3 ?
 (1) 81 ml (2) 60 ml
 (3) 71 ml (4) 52 ml

ANSWER KEY

1.	(1)	2.	(2)	3.	(4)	4.	(4)	5.	(3)	6.	(2)	7.	(4)	8.	(4)	9.	(3)	10.	(3)
11.	(2)	12.	(1)	13.	(4)	14.	(2)	15.	(2)	16.	(2)	17.	(1)	18.	(3)	19.	(4)	20.	(1)
21.	(2)	22.	(4)	23.	(4)	24.	(3)	25.	(1)	26.	(4)	27.	(4)	28.	(3)	29.	(3)	30.	(1)

CDS PYQ

1. In a certain school, the ratio of boys to girls is 7:5. If there are 2400 students in the school, then how many girls are there?
[CDS 2013 (I)]
(a) 500 (b) 700
(c) 800 (d) 1000
2. If a, b, c, d and e are in continued proportion then $a/e =$
[CDS 2013 (I)]
(a) a^3/b^3 (b) a^4/b^4
(c) b^3/a^3 (d) b^4/a^4
3. The present age of Ravi's father is 4 times of Ravi's present age. 5 years back Ravi father was 7 times as old as Ravi was at that time. What is the present age of Ravi's father?
[CDS 2014 (I)]
(a) 84 years (b) 70 years
(c) 40 years (d) 35 years
4. If $A:B = 2:3$, $B:C = 5:7$ and $C:D = 3:10$, then what is A: D equal to?
[CDS 2014 (I)]
(a) 1:7 (b) 2:7
(c) 1:5 (d) 5:1
5. $(x+y) : (x-y) = 3:5$ and $xy =$ positive imply that
[CDS 2014 (II)]
(a) x and y are both positive
(b) x and y are both negative
(c) one of them is positive and one of them is negative
(d) no real solution for x and y exists
6. The ratio of ages of A and B is 2:5 and the ratio of ages of B and C is 3:4. What is the ratio of ages of A, B and C?
[CDS 2014 (II)]
(a) 6 : 15 : 20 (b) 8 : 5 : 3
(c) 6 : 5 : 4 (d) 2 : 15 : 4
7. The height of a tree varies as the square root of its age (between 5 to 17 yr). When the age of the tree is 9 yr, its height is 4 ft. What will be the height of the tree at the age of 16 yr?
[CDS 2014 (II)]
(a) 5 ft 4 inch (b) 5 ft 5 inch
(c) 4 ft 4 inch (d) 4 ft 5 inch
8. 16 L of a mixture contains milk and water in the ratio 5:3. If 4 L of milk is added to this mixture, the ratio of milk to water in the new mixture would be
[CDS 2015 (I)]
(a) 2: 1 (b) 7:3
(c) 4:3 (d) 8:3
9. If $a:b=3:5$ and $b:c=7:8$, then $2a: 3b: 7c$ is equal to
[CDS 2015 (II)]
(a) 42: 105: 320 (b) 15: 21: 40
(c) 6:15: 40 (d) 30 21 350
10. In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7: 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1: 2 is
[CDS 2015 (II)]
(a) 30 (b) 32
(c) 33 (d) 35
11. The annual incomes of two persons are in the ratio 9:7 and their expenses are in the ratio 4:3. If each of them saves Rs 2000 per year, what is the difference in their annual incomes?
[CDS 2016 (I)]
(a) Rs 4000 (b) Rs 4500
(c) Rs 5000 (d) Rs 5500
12. If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, then which of the following is/are correct ?
I. $\frac{b^3+c^3+d^3}{a^3+b^3+c^3} = \frac{d}{a}$
II. $\frac{a^2+b^2+c^2}{b^2+c^2+d^2} = \frac{a}{d}$
Select the correct answer using the code given below
[CDS 2016 (I)]
(a) Only 1 (b) Only 2
(c) Both I and II (d) Neither I nor II
13. In an office, one third of the workers are women, half of the women are married and one-third of the married women have children, If three-fourth of the men are married and one third of married men have children, then what is the ratio of married women to married men ?
[CDS 2016 (II)]
(a) 1:2 (b) 2:1
(c) 3:1 (d) 1:3

14. There are twelve friends A, B, C, D, E, F, G, H, I, J, K and L who invested money in some business in the ratio of 1 : 2 : 3 : 4 : 5 : 6 : 7 : 8 : 9 : 10 : 11 : 12 and the duration for which they invested the money is in the ratio of 12 : 11 : 10 : 9 : 8 : 7 : 6 : 5 : 4 : 3 : 2 : 1 respectively. Who will get the maximum profit at the end of the year ?

[CDS 2016 (II)]

- (a) F only (b) G only
(c) Both F and G (d) Neither F nor G

15. Income of Mahesh and Kamal are in the ratio 1:2, and their expenses are in the ratio 1:3, Which one of the following statements are correct ?

[CDS 2016 (II)]

- (a) Mahesh saves more than what Kamal saves
(b) Saving of both of them are equal
(c) Kamal save more than what Mahesh saves
(d) It is not possible to determine who saves more

16. X and Y entered into partnership with Rs. 700 and Rs 600 respectively. After 3 months X withdraw $\frac{2}{7}$ of his stock but after 3 months, he puts back $\frac{3}{5}$ of what he had drawn. The profit at the end of the year is Rs. 726. How much of this should X receive ?

[CDS 2016 (II)]

- (a) Rs 336 (b) Rs 366
(c) Rs 633 (d) Rs 663

17. Leela got married 6 years ago. Today her age is $1\frac{1}{4}$ times her age at the time of her

marriage. Her son's age is $\frac{1}{10}$ times her age.

What is the present age of her son?

[CDS 2017 (I)]

- (a) 1 years (b) 2 years
(c) 3 years (d) 4 years

18. The ratio of two numbers is 1 : 5 and their product is 320. What is the difference between the squares of these two numbers?

[CDS 2017 (I)]

- (a) 1024 (b) 1256
(c) 1536 (d) 1640

19. 25 kg of alloy X is mixed with 125 of alloy Y. If the amount of lead and tin in the alloy X is in the ratio 1 : 2 and the amount of lead and

tin in the alloy Y is in the ratio 2 : 3, then what is the ratio of lead to tin in the mixture?

[CDS 2017 (I)]

- (a) 1 : 2 (b) 2 : 3
(c) 3 : 5 (d) 7 : 11

20. The cost of a diamond varies directly as the square of its weight. A diamond broke into four pieces with their weights in the ratio of 1:2:3:4. If the loss in total value of the diamond was Rs.70,000, what was the price of the original diamond?

[CDS 2017 (I)]

- (a) Rs 1,00,000 (b) Rs 1,40,000
(c) Rs 1,50,000 (d) Rs 1,75,000

21. In a class of 49 students, the ratio of girls to boys is 4:3. If 4 girls leave the class, the ratio of girls to boys would be:

[CDS 2017 (I)]

- (a) 11:7 (b) 8:7
(c) 6:5 (d) 9:8

22. Five year ago Ram was three times as old as Shyam. Four years from now, Ram will be only twice as old as Shyam what is the percentage of Ram?

[CDS 2017 (II)]

- (a) 30 years (b) 32 years
(c) 36 years (d) 40 years

23. If 78 is divided into 3 parts which are proportional to $1, \frac{1}{3}, \frac{1}{6}$, then what is the middle part

[CDS 2017 (II)]

- (a) $28/3$ (b) 13
(c) $52/3$ (d) $56/3$

24. There are 350 boys in the first three standards. The ratio of the number of boys in first and second standards is 2:3, while that of boys in second and third standards is 4:5. What is the total number of boys in first and third standards?

[CDS 2017 (II)]

- (a) 302 (b) 280
(c) 242 (d) 230

25. If A:B = 1:2, B:C = 3:4, C:D = 2:3 and D:E = 3:4, then what is B:E equal to?

[CDS 2018 (I)]

- (a) 3:2 (b) 1:8
(c) 3:8 (d) 4:1

26. Rs.120 are divided among A,B,C such that A's share is Rs. 20 more than B's and Rs.20 less than C's. What is B's share?

[CDS 2018 (I)]

- (a) ₹10 (b) ₹15
(c) ₹20 (d) ₹25

27. A, B, C, D and E start a partnership firm. Capital contributed by A is three times that contributed by D. E contributes half of A's contribution, B contributes one third of E's contribution and C contributes two third of A's contribution. If the difference between the combined shares of A, D and E and the combined shares of B and C in the total profit of the firm is Rs. 13500, what is the combined share of B, C and E? (The shares are supposed to be proportional to the contributions)

[CDS 2018 (I)]

- (a) Rs. 135000 (b) Rs. 18000
(c) Rs. 19750 (d) Rs. 20250

28. An alloy A contains two elements, copper and tin in the ratio of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3:4. If 20 kg of alloy A, 28 kg of alloy B and some more pure copper are mixed to form a third alloy C which now contains copper and tin in the ratio of 6:7, then what is the quantity of pure copper mixed in the alloy C?

[CDS 2018 (I)]

- (a) 3kg (b) 4kg
(c) 5kg (d) 7kg

29. The sum of ages of a father a mother, a son Sonu and daughter Savita and Sonia is 96 years. Son is the youngest member of the family. The year Sonu was born, the sum of the ages of all the members of the family was 66 years. If the father's age now is six times that of Sonu's present age ,then 12 years hence the father's age will be

[CDS 2018 (II)]

- (a) 44 years (b) 45 years
(c) 46 years (d) 48 years

30. 10 years before the age of a mother and her daughter were in the ratio 3:1, in another 10 years from now the ratio of their ages will be 13:7. what are their present age

[CDS 2018 (II)]

- (a) 39 yr, 21 yr (b) 55 yr, 25 yr
(c) 75 yr, 25 yr (d) 49 yr, 31 yr

31. The ratio of the sum and difference of the ages of the father and the son is 11:3, Consider the following statements

1. The ratio of their age is 8 : 5
 2. The ratio of their ages after the son attains twice the present age will be 11:8
- Which of the statements given above are correct?

[CDS 2018 (II)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

32. If $a:b = c:d = 1:6$, then what is the value of $\frac{a^2 + c^2}{b^2 + d^2}$?

[CDS 2019 (I)]

- (a) 1/600 (b) 1/60
(c) 1/36 (d) 1/6

33. If $(4a+7b)(4c-7d)=(4a-7b)(4c+7d)$, then which one of the following is correct ?

[CDS 2019 (I)]

- (a) $\frac{a}{b} = \frac{c}{d}$ (b) $\frac{a}{d} = \frac{c}{b}$
(c) $\frac{a}{b} = \frac{d}{c}$ (d) $\frac{4a}{7b} = \frac{c}{d}$

34. Priya's age was cube of an integral number (different from 1) four years ago and square of an integral number after four years. How long should she wait so that her age becomes square of a number in the previous year and cube of a number in the next year?

[CDS 2019 (I)]

- (a) 7yr (b) 12 yr
(c) 14 yr (d) 21 yr

35. The monthly incomes of A and B are in the ratio 4:3. Each saves 600. If their expenditures are in the ratio 3 :2, then what is the monthly income of A

[CDS 2019 (II)]

- (a) 1800 (b) 2000
(c) 2400 (d) 3600

36. Radha and Rani are sisters, five years back the age of Radha was 3 times that of Rani, but one year back the age of Radha was 2 times that of Rani. What is age difference between them:

[CDS 2019 (II)]

- (a) 8 (b) 9
(c) 10 (d) 11

37. The population of two villages is 1525 and 2600 respectively. If the ratio of male to female population in the first village is 27:34 and the ratio of male to female population in the second village is 6:7, then what is the ratio of male to female population of these two villages taken together?

[CDS 2020 (I)]

- (a) $\frac{33}{41}$ (b) $\frac{85}{82}$
(c) $\frac{71}{90}$ (d) $\frac{5}{6}$

38. If $\frac{a}{b} = \frac{1}{3}$, $\frac{b}{c} = 2$, $\frac{c}{d} = \frac{1}{2}$, $\frac{d}{e} = 3$ and $\frac{e}{f} = \frac{1}{4}$,

then what is the value of $\frac{abc}{def}$?

[CDS 2020 (I)]

- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) $\frac{3}{8}$ (d) $\frac{27}{4}$

39. Three persons start a business with capitals in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{5}$. The first person withdraws half his capital after 4 months. What is the share of profit of A if the business fetches an annual profit of Rs 96800?

[CDS 2020 (II)]

- (a) Rs 32000 (b) Rs 34500
(c) Rs 36000 (d) Rs 36800

40. Ena was born 4 years after her parents marriage. Her mother is 3 years younger than her father and 24 years older than Era, who is 13 years old. At what age did Rea's father get married?

[CDS 2020 (II)]

- (a) 25 years (b) 24 years
(c) 23 years (d) 22 years

41. Mahesh is 60 years old. Ram is 5 years younger to Mahesh and 4 years elder to Raju. Babu is younger brother of Raju and he is 6 years younger. What is age difference between Mahesh and Babu?

[CDS 2020 (II)]

- (a) 18 years (b) 15 years
(c) 13 years (d) 11 years

42. The income of A, B and C are in the ratio 7 : 9 : 12 and their expenditures are in the ratio 8 : 9 : 15. If A's saving is one-fourth of his income, then the ratio of savings of A, B and C is

[CDS 2021 (I)]

- (a) 56 : 99 : 69 (b) 99 : 56 : 69
(c) 69 : 56 : 99 (d) 99 : 69 : 56

43. If three positive numbers are in the ratio 2 : 3 : 5 and the sum of their squares is 1368, then what is the sum of the numbers?

[CDS 2021 (II)]

- (a) 30 (b) 45
(c) 60 (d) 75

44. If $a = 3b$, $4b = 5c$, $6c = 7d$, then what is

$\frac{d+a}{d-a}$ equal to?

[CDS 2022 (II)]

- (a) $\frac{27}{43}$ (b) $\frac{43}{27}$
(c) $-\frac{27}{43}$ (d) $-\frac{43}{27}$

45. Two glasses of equal volume are filled with a mixture of alcohol and water in the ratio 3:2 and 4:1, respectively. These glasses are emptied into a third glass. What is the ratio of alcohol and water in the third glass?

[CDS 2022 (II)]

- (a) 5:4 (b) 7:2
(c) 7:3 (d) 7:4

46. A, B, C, D and E enter into a business. They invest money in the ratio 2 : 3 : 4 : 5 : 6. However, the time invested by them is in the ratio 6 : 5 : 4 : 3 : 2. If the profit distributed is directly proportional to time and money invested, then who receives the highest amount of profit?

[CDS - 2023 (1)]

- (a) C
(b) both B and D
(c) both c and D
(d) All get equal profit

47. If a, b, c, d, e and f satisfy $2a = 3b = 6c = 9d = 12e = 18f$, then what is the value of $(a + b)/(c + d + e + f)$?

[CDS - 2023 (1)]

- (a) $\frac{4}{7}$ (b) 2
(c) $\frac{5}{2}$ (d) $\frac{9}{2}$

48. Three glasses P, Q and R have capacities in the ratio 1:2:3. All these glasses are completely filled with mixtures of milk and water. The ratio of milk to water in P is 1:2, in Q it is 2:3 and in R it is 3:1. If the content

of all three glasses are put into a bigger container, what will be the ratio of milk to water in the container?

[CDS – 2023 (2)]

- (a) 203:117 (b) 203:157
(c) 172:91 (d) 165:88

49. Rs 9400 is distributed among P, Q, R in such a way that if Rs 93, Rs 24, Rs 55 are deducted from their respective shares, then they have money in the ratio 3 : 4 : 5. What is the share of P?

[CDS – 2024 (1)]

- (a) Rs 2307 (b) Rs 2376
(c) Rs 2508 (d) Rs 12896

50. If $(a + b) : (b + c) : (c + a) = 5 : 7 : 6$, then what is the value of $(a - b + c) : (a + b - c)$?

[CDS – 2024 (1)]

- (a) 1 : 1 (b) 2 : 3
(c) 3 : 1 (d) 4 : 3

51. A bottle contains spirit and water in the ratio 1 : 4 and another identical bottle contains spirit and water in the ratio 4 : 1. In what ratio should the mixtures in the two bottles be mixed to get a new mixture in which the ratio of spirit to water is 1 : 3?

[CDS – 2024 (1)]

- (a) 5 : 1 (b) 6 : 1
(c) 10 : 1 (d) 11 : 1

52. If $a : b : c : d = \sqrt{4} : \sqrt{3} : \sqrt{2} : \sqrt{1}$, then what is the value of $\frac{(-a^2 + b^2 + c^2 + d^2)}{(a^2 - b^2 + c^2 - d^2)}$?

[CDS – 2024 (1)]

- (a) 1 (b) 2
(c) 3 (d) 6

53. If $m : n = 1 : 2$ and $p : q = 3 : 4$, then what is $(2m + 4p) : (n + 3q)$ equal to?

[CDS – 2024 (1)]

- (a) 1 : 1 (b) 1 : 3
(c) 2 : 1 (d) 2 : 3

ANSWER KEY

1.	d	2.	b	3.	c	4.	a	5.	d	6.	a	7.	a	8.	b	9.	c	10.	c
11.	a	12.	a	13.	d	14.	c	15.	d	16.	b	17.	c	18.	c	19.	d	20.	a
21.	b	22.	b	23.	c	24.	d	25.	c	26.	c	27.	b	28.	b	29.	d	30.	b
31.	b	32.	c	33.	a	34.	c	35.	c	36.	a	37.	d	38.	c	39.	a	40.	c
41.	b	42.	a	43.	c	44.	d	45.	c	46.	c	47.	b	48.	b	49.	a	50.	c
51.	d	52.	a	53.	a														

CAPF PYQ

1. 5 % of income of A is equal to 15% of income of B and 10% income of B is equal to 20% of income of C. If the income of C is Rs. 2000, then what is the total income of A,B and C is
[CAPF 2016]
(a) Rs. 12000 (b) Rs. 18000
(c) Rs. 14000 (d) Rs. 6000
2. One year ago, a father was four times as old as his son. After six years his age exceeds twice his son's age by 9 years. The ratio of their present age is
[CAPF 2018]
(a) 9 : 2 (b) 11 : 3
(c) 12 : 5 (d) 13 : 4
3. The ratio of ages of a man and his son is 3 : 1. After 15 years, the age ratio will be 2 : 1, What is the age of man ?
[CAPF 2018]
(a) 45 years (b) 40 years
(c) 35 years (d) 30 years
4. If 15% of A is double of 30% of B, then what is the ratio of A to B ?
[CAPF 2020]
- (a) 1 : 2 (b) 2: 1
(c) 1 : 4 (d) 4 : 1
5. Two years ago , the age of A was three times the age of B , If B is currently 9 years old , then after how many years The age of A will be double of the age of B ?
[CAPF 2020]
(a) 2 years (b) 3 years
(c) 4 years (d) 5 years
6. The ratio of monthly income of A and B is 7:10. The ratio of their expenditures is 2:3. If each of A and B saves Rs 1000 per month, then what will be the monthly income of B?
[CAPF 2021]
(a) Rs 9000 (b) Rs 10000
(c) Rs 15000 (d) Rs 12000
7. A, B and C start a business by investing Rs 7000, Rs 8000 and Rs 12000 respectively. After a year, B get Rs 3200 as his share of profit. What is the total profit?
[CAPF 2021]
(a) Rs 16600 (b) Rs 10000
(c) Rs 21600 (d) Rs 10800

ANSWER KEY

1.	(b)	2.	(b)	3.	(a)	4.	(d)	5.	(d)	6.	(b)	7.	(d)						
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